Approximate Multiply–Accumulate Units for Energy-Efficient DNN Accelerators

Error Characterization and Application-Level Evaluation

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Abstract

Multiply–accumulate (MAC) operations dominate both the computation and the energy budget of deep neural network (DNN) inference. Because DNNs are inherently tolerant to small numerical errors, the exact multiplier inside each MAC can be replaced with a cheaper *approximate* multiplier. This project studies that trade-off for 8-bit signed arithmetic. Bit-accurate models of an exact array multiplier, three column-truncated multipliers, and Mitchell's logarithmic multiplier were developed and characterized exhaustively over all 2^{16} input pairs. A fully-connected network was trained on MNIST, quantized to int8 with int32 accumulation, and evaluated with every multiplication routed through each approximate design. Results show that even a multiplier with a mean relative error of 23.9 % reduces classification accuracy by only 0.42 %, confirming that large hardware savings are available at negligible application-level cost.

1 Introduction

The processing elements (PEs) of a DNN accelerator are, at their core, MAC units. In an 8-bit datapath the multiplier is the largest and most power-hungry part of the PE, so its cost scales directly with the size of the PE array. *Approximate computing* exploits the error resilience of DNNs: errors in individual products are partially averaged out across long dot products, and the final classification decision (an `argmax`) is insensitive to small perturbations of the output scores.

The goals of this project are:

1. to build bit-accurate, hardware-oriented models of representative approximate multipliers;
2. to characterize their arithmetic error with standard metrics;
3. to measure their impact on the accuracy of a quantized neural network;
4. to implement them in Verilog and quantify the hardware savings (in progress, Section 5).

2 Multiplier Designs

All designs operate on 8-bit two's-complement operands $a, b \in [-128, 127]$ and are implemented in sign-magnitude form: the magnitudes $|a|, |b| \in [0, 128]$ are multiplied and the sign is restored as $\mathrm{sgn}(a) \oplus \mathrm{sgn}(b)$. This makes the error of every approximate design symmetric around zero.

2.1 Exact array multiplier (baseline)

The product is the sum of N^2 partial-product bits,

$$P = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} a_i \, b_j \, 2^{i+j}, \qquad N = 8. \tag{1}$$

2.2 Column-truncated multiplier

All partial-product bits whose weight falls below column k are removed:

$$P_{\text{trunc}}(k) = \sum_{i+j \,\geq\, k} a_i \, b_j \, 2^{i+j}. \tag{2}$$

Each removed bit eliminates one AND gate and its share of the carry-save adder tree. For $k = 4, 6, 8$, respectively 10, 21 and 36 of the 64 partial-product bits are removed. No error-compensation constant is added, so the design always under-estimates the product magnitude.

2.3 Mitchell logarithmic multiplier

Writing each operand as $x = 2^k(1 + f)$, where $k = \lfloor \log_2 x \rfloor$ is found by a leading-one detector (LOD) and $0 \leq f < 1$, Mitchell's algorithm [1] approximates $\log_2(ab) \approx k_1 + k_2 + f_1 + f_2$. With $x_i = |a_i| - 2^{k_i}$, an exact-integer form suitable for hardware is

$$S = x_1 2^{k_2} + x_2 2^{k_1}, \qquad P_{\text{Mitchell}} = \begin{cases} 2^{k_1+k_2} + S, & S < 2^{k_1+k_2} \\ 2S, & \text{otherwise.} \end{cases} \tag{3}$$

The datapath therefore consists only of two LODs, barrel shifters and a single adder; no partial-product array is required. The design always under-estimates, with a worst-case relative error of about 11.1 %.

3 Methodology

3.1 Bit-accurate reference models

Each design is described in Python/NumPy at the bit level (`python/approx_mult.py`), mirroring the intended RTL structure (partial-product selection, LOD, shifts). These models serve as the golden reference for RTL verification. For fast evaluation, each model is expanded into a 256×256 product look-up table.

3.2 Error metrics

Every design is evaluated over all 65,536 input pairs. With P the exact and $\hat{P}$ the approximate product, and $\text{ED} = |\hat{P} - P|$, the following metrics are reported [2]:

- **ER** – error rate, fraction of inputs with ED $\neq 0$;
- **MED** – mean error distance, $\mathbb{E}[\text{ED}]$;
- **NMED** – MED normalized by the maximum product (128^2);
- **MRED** – mean relative error distance, $\mathbb{E}[\text{ED}/|P|]$ over $P \neq 0$;
- **Max ED** – worst-case absolute error.

3.3 Quantized neural network

A 784–128–10 multilayer perceptron with ReLU activation was trained on MNIST [5] for eight epochs using mini-batch SGD (batch 128, learning rate 0.1), implemented directly in NumPy. The trained network was then quantized following the integer-only inference scheme of [4]:

- inputs: $x_q = \text{round}(127\,x)$, $x \in [0, 1]$;
- weights: symmetric per-tensor int8, $s_w = \max|W|/127$;
- biases: int32 at scale $s_x s_w$;
- accumulation: int32;
- hidden activations: requantized to unsigned 7-bit values using a scale calibrated on 5,000 training images (99.99th percentile).

During approximate inference, *every* multiplication in both layers is replaced by the corresponding approximate multiplier, exactly as it would occur inside the accelerator's PEs. Accuracy is measured on the full 10,000-image test set.

4 Results

4.1 Arithmetic error

Table 1 summarizes the exhaustive error characterization, and Figure 1 shows how the relative error is distributed over the input space.

Table 1: Error characterization over all 2^{16} signed 8-bit input pairs.

Multiplier	ER (%)	MED	NMED (%)	MRED (%)	Max ED
Exact	0.00	0.00	0.000	0.00	0
Truncated k=4	81.25	12.25	0.075	1.72	49
Truncated k=6	93.75	80.25	0.490	7.59	321
Truncated k=8	98.05	384.75	2.348	23.87	1537
Mitchell	87.89	151.34	0.924	3.70	1024

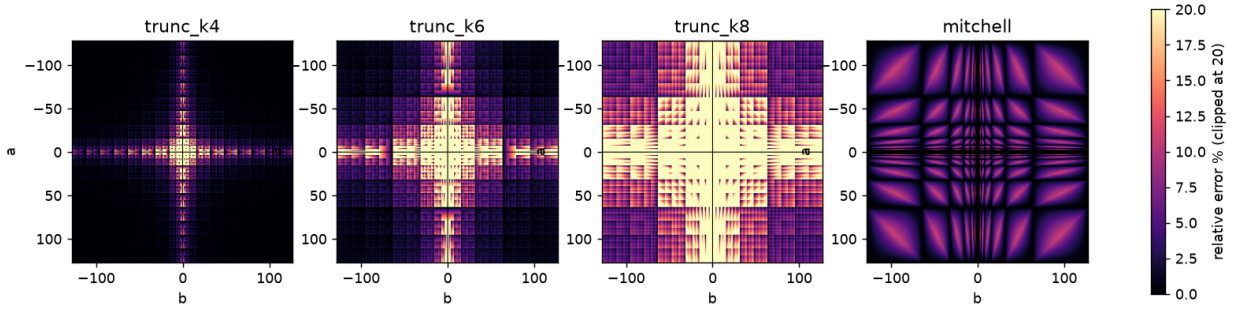


Figure 1: Relative error (%) of each approximate multiplier as a function of its operands. Truncation errors concentrate at small operands, where the removed low-order columns carry a larger share of the product; Mitchell's error depends on the fractional parts of both operands and is bounded at about 11 %.

The measured MRED of the Mitchell multiplier (3.70 %) agrees with the value reported in the literature (≈3.8 %), which validates the model. Truncation exhibits a clear knob: each increase of k by two columns raises the MED by roughly a factor of five.

4.2 Application-level accuracy

Table 2: MNIST test accuracy of the int8 MLP with each multiplier. Float32 reference: 96.82 %.

Multiplier	Accuracy (%)	Δ vs. exact int8
Exact (int8)	96.81	–
Truncated k=4	96.82	+0.01
Truncated k=6	96.77	−0.04
Truncated k=8	96.39	−0.42
Mitchell	96.77	−0.04

Table 2 shows the central result of this project: arithmetic error and application error are only weakly coupled. Int8 quantization itself costs just 0.01 %. Truncation up to $k = 6$ and the Mitchell multiplier are practically lossless, and even $k = 8$ truncation—which removes more than half of the partial-product array and has an MRED above 20 %—loses less than half a percentage point. Two effects explain this resilience:

1. **Error averaging.** Each output neuron accumulates up to 784 products; independent errors partially cancel, so the relative error of the sum is far smaller than that of a single product.
2. **Decision robustness.** Classification depends only on the ranking of the output scores, so a systematic, roughly proportional under-estimation (as produced by both truncation and Mitchell) rarely changes the `argmax`.

5 Work in Progress

1. **RTL implementation.** Parameterized Verilog modules for the exact, truncated and Mitchell multipliers, and a MAC wrapper with an int32 accumulator.
2. **Exhaustive RTL verification.** A self-checking testbench that compares all 2^{16} products of each design against the Python golden model.
3. **Hardware cost.** Logic synthesis with Yosys to compare cell count and estimated area, giving the final *area saving vs. accuracy loss* Pareto front.
4. **Extensions.** Error-compensated truncation, and evaluation on a convolutional network with a harder dataset.

6 Repository Structure

```
python/approx_mult.py bit-accurate multiplier models
python/error_analysis.py exhaustive error metrics and heatmaps
python/train_quantize.py NumPy MLP training and int8 quantization
python/approx_inference.py inference through approximate multipliers
results/ metrics (CSV), figures, quantized model
docs/report.tex this report
```

All experiments are reproducible with `pip install -r requirements.txt` followed by running the four scripts in the order listed above.

References

[1] J. N. Mitchell, "Computer multiplication and division using binary logarithms," *IRE Transactions on Electronic Computers*, vol. EC-11, no. 4, pp. 512–517, 1962.

[2] H. Jiang, F. J. H. Santiago, H. Mo, L. Liu, and J. Han, "Approximate arithmetic circuits: A survey, characterization, and recent applications," *Proceedings of the IEEE*, vol. 108, no. 12, pp. 2108–2135, 2020.

[3] V. Mrazek, R. Hrbacek, Z. Vasicek, and L. Sekanina, "EvoApprox8b: Library of approximate adders and multipliers for circuit design and benchmarking of approximation methods," in *Proc. DATE*, 2017, pp. 258–261.

[4] B. Jacob *et al.*, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. CVPR*, 2018, pp. 2704–2713.

[5] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.